

Lab01 – Containerize an application

In this exercise, you will practice building a container image from an existing `Dockerfile`. Then you will run the container from the image, and interact with it.

1. Inspect the `Dockerfile` (./app/Dockerfile) in the `app` directory.

The `Dockerfile` builds a nodejs-based application. All files necessary to run the application are available in the same directory. The `Dockerfile` exposes port 3000.

```
brahim@Training:~/lab01$ tree
.
└── app
    ├── Dockerfile
    ├── index.js
    └── package.json

1 directory, 3 files
brahim@Training:~/lab01$
brahim@Training:~/lab01$ cat app/Dockerfile
FROM node:20.1.0-alpine
WORKDIR /app
COPY ["package.json", "package-lock.json*", "./*"]
RUN npm install --production
COPY .
EXPOSE 3000
CMD ["node", "index.js"]

brahim@Training:~/lab01$
brahim@Training:~/lab01$
```

2. Build the container image from the `Dockerfile` with the tag `nodejs-hello-world:1.0.0`.

```
brahim@Training:~/lab01$ docker build -t nodejs-hello-world:1.0.0 app/
[+] Building 122.0s (12/12) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 202B
=> [internal] load .dockerignore
=> => transferring context: 66B
=> [internal] load metadata for docker.io/library/node:20.1.0-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [1/5] FROM docker.io/library/node:20.1.0-alpine@sha256:6e56967f8a4032f084856bad4185088711d25b2c2c705af84f57a522c84d123b
=> => resolve docker.io/library/node:20.1.0-alpine@sha256:6e56967f8a4032f084856bad4185088711d25b2c2c705af84f57a522c84d123b
=> => sha256:6e56967f8a4032f084856bad4185088711d25b2c2c705af84f57a522c84d123b 1.43kB / 1.43kB
=> => sha256:71073d3b7ad12dc871ecf42a865699914efd738fe797fa16f67a22179f46d039 1.16kB / 1.16kB
=> => sha256:8e7579c71aa85206d1fa454670cf4b5a847f54ac27077136a288f64e6690e37c 6.48kB / 6.48kB
=> => sha256:3f026796e5ad2d5022169e508286b0380f6a15ce0340dd744ddae178df49982c 49.55MB / 49.55MB
=> => sha256:08556a236d1cb089410b2fc0f20b00d89a47698bda3e48a28eb7c582a942158e 2.35MB / 2.35MB
=> => sha256:06e990ca428a9474589c4e5c569a40e437327d58d46a0c785f56be6db425485 451B / 451B
=> => extracting sha256:3f026796e5ad2d5022169e508286b0380f6a15ce0340dd744ddae178df49982c
=> => extracting sha256:08556a236d1cb089410b2fc0f20b00d89a47698bda3e48a28eb7c582a942158e
=> [internal] load build context
=> => transferring context: 880B
=> [auth] library/node:pull token for registry-1.docker.io
=> [2/5] WORKDIR /app
=> [3/5] COPY [package.json, package-lock.json*, ./*]
=> [4/5] RUN npm install --production
=> [5/5] COPY .
=> exporting to image
=> => exporting layers
=> => writing image sha256:4c2d39df03509ea4616ba3df3824f7f1e3c7ddf8123f51e7ba8d6c2ebb80db6c
=> => naming to docker.io/library/nodejs-hello-world:1.0.0
brahim@Training:~/lab01$ docker images nodejs-hello-world:1.0.0
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
nodejs-hello-world  1.0.0           4c2d39df0350   39 seconds ago 180MB
brahim@Training:~/lab01$
```

- Run a container with the container image. Make the application available on port 80.

```
brahim@Training:~/Lab01$ docker run -d -p 80:3000 nodejs-hello-world:1.0.0
f56558b5a889f84456167d8feaf5434a6993cda12568b8095c5319cf7961af8

brahim@Training:~/Lab01$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
f56558b5a889   nodejs-hello-world:1.0.0   "docker-entrypoint.s..."  28 seconds ago Up 6 seconds   0.0.0.0:80->3000/tcp, :::80->3000/tcp   strange_moore

brahim@Training:~/Lab01$ docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
f56558b5a889   nodejs-hello-world:1.0.0   "docker-entrypoint.s..."  36 seconds ago Up 14 seconds   0.0.0.0:80->3000/tcp, :::80->3000/tcp   strange_moore
brahim@Training:~/Lab01$
```

- Use your browser or Execute a `curl` or `wget` command against the application's endpoint.



- Retrieve the container logs.

```
brahim@Training:~/Lab01$ docker logs strange_moore
Magic happens on port 3000
brahim@Training:~/Lab01$
brahim@Training:~/Lab01$
```

- Save the container image to the file `nodejs-hello-world-1.0.0.tar`.

```
brahim@Training:~/Lab01$ docker save -o nodejs-hello-world-1.0.0.tar nodejs-hello-world:1.0.0
brahim@Training:~/Lab01$
brahim@Training:~/Lab01$ ls -l nodejs-hello-world-1.0.0.tar
-rw-r--r-- 1 brahim brahim 184518144 juil. 25 13:41 nodejs-hello-world-1.0.0.tar
brahim@Training:~/Lab01$
brahim@Training:~/Lab01$
```